PROVISIONAL APPLICATION FOR PATENT

UNDER 35 U.S.C. § 111(b)

SAFETY GUARD STATE RESPONSIVE TO A SAFE PHRASE ARTICULATED SILENTLY AND SENSED WITHOUT A MICROPHONE; SILENT CHECK-IN ANSWERS AND CONSENT SIGNALS; NATURAL-LANGUAGE SAFETY INSTRUCTIONS; AND WEARABLE-BUTTON AND AI-ASSISTANT SAFETY INPUTS

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Related applications: U.S. Provisional Patent Application No. 64/183,657; U.S. Provisional Patent Application No. 64/189,542; U.S. Patent No. 12,347,301 B1; U.S. Provisional Patent Application No. 64/016,800; U.S. Patent Application Publication No. 2016/0196735 A1 (each incorporated by reference)

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CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is related to U.S. Provisional Patent Application No. 64/183,657, filed October 24, 2026, titled "Privacy-Preserving Ambient Consent Signaling Gated on Continuously Renewed Affirmation, with Music, Lapse Detection and Escalation to Help" (the "'657 application"), and to U.S. Provisional Patent Application No. 64/189,542, filed November 2, 2026, titled "Bedroom Safety Guard State Armed by an Ordinary Cue, with Wake-Word-Free Safe-Phrase Listening and Private Escalation; Arrival-Triggered Safety Check-In; On-Demand Refuge Lodging; a Verifiable Mediated Apology Channel; and Privacy-Preserving Distancing" (the "'542 application"), each of the present inventor, the entire disclosure of each of which is incorporated herein by reference. Terms defined in those applications have the same meanings here unless this application defines them differently. This application is nonetheless written to stand on its own: the guard state, private check-in, help response, and affirmative consent envelope with which the inputs

described below are used are restated in section 2 of the detailed description, and every input described below may be used together with, or independently of, any mechanism of those applications.

[0002] This application is also related to U.S. Patent No. 12,347,301 B1, titled "Consent management system over a communications network," issued to the present inventor (the "'301 patent"); to U.S. Provisional Patent Application No. 64/016,800, titled "Networked Emergency Dispatch and Preparedness Simulation System"; and to U.S. Patent Application Publication No. 2016/0196735 A1 of the present inventor (published under the name Adam Clayman), titled "Systems and Methods for Monitoring Health in a Shared Living Environment." Each is incorporated herein by reference in its entirety.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0003] Not applicable.

TECHNICAL FIELD

[0004] This disclosure relates to personal-safety and consent-signaling systems that run on phones, wearable devices, smart speakers, and connected media devices. More particularly, it relates to detecting a safe phrase, an answer to a check-in, an arming cue, or an affirmation or stop signal that a person articulates silently, whispers, or speaks, from the signal of a non-acoustic sensor of the person's articulators, such as a radio-frequency sensor in the frames of eyewear; to inputs of a wearable button that opens a voice session while it is held; and to an interface through which an artificial-intelligence assistant invokes such a system.

BACKGROUND

[0005] A safe word works only if it can be said. Systems that listen for a spoken safe phrase, including those of the related applications, rely on a microphone, but a person in danger is often unable to speak aloud. Fear can produce a freeze response, sometimes called tonic immobility, in which a person cannot move or call out; a hand, a pillow, or another person's weight can muffle the voice; and a person may judge that speaking aloud would provoke the other person. A phrase spoken aloud is also heard by the very person whose conduct prompted it.

[0006] A microphone in a bedroom raises privacy concerns of its own, among them capturing the speech of a person who has not agreed to be recorded. Its accuracy also suffers while music or other

sound plays, which is why the related applications cancel or otherwise suppress media so that a song's lyrics are not mistaken for a safe phrase.

[0007] Silent-speech interfaces recognize words that a person articulates with little or no audible sound. Research systems and products have used surface electromyography of the facial and neck muscles, inertial and vibration sensing of the jaw and throat, ultrasonic and camera-based sensing of the lips, and radio-frequency sensing, in which antennas, for example in the frames of eyeglasses, pass low-power radio signals through the face and infer the positions of the tongue, lips, and soft palate from the received signal. Such interfaces have been offered for text entry and for commands to computers and assistants.

[0008] Separately, small wearable and clip-on buttons offer press, double-press, hold, and directional-swipe inputs and, in some products, a microphone that is active only while a button is held, so that holding the button opens a voice session with an assistant whose answer plays on a paired phone or headphones. Smart buttons have also been used to send personal-safety alerts.

[0009] A need exists for safety and consent systems that use these inputs while keeping their advantages (silence, attribution to the person wearing the sensor, and freedom from any room microphone) and avoiding their risks, such as a silent signal that the other person cannot perceive, a wearable device that is taken off before bed, or a safety function that depends on a particular assistant or its provider.

SUMMARY

[0010] In one aspect, a device of a user enters a guard state when the user gives an arming cue, which may be an ordinary input that names no safety function, such as a song, a routine request to a voice assistant, a button pattern, or a phrase that the user articulates silently. In the guard state, the device detects, without first detecting a wake word, a safe phrase that the user articulates aloud, in a whisper, or silently, from the signal of a non-acoustic sensor of the movement or activity of the user's articulators, such as a radio-frequency sensor in the frames of eyewear, alone or fused with a microphone. Upon the safe phrase, the device presents a private check-in on a device of the user and, absent a response, initiates a tiered help response.

[0011] Because such a sensor senses only the person who wears it, a silently articulated safe phrase cannot be given by another person, is not triggered by a song's lyrics or a television, and needs no cancellation of media; and because nothing is said aloud, the other person hears nothing. The guard state can run with no microphone open in the room, so that no one's speech is captured.

[0012] In another aspect, the user answers the private check-in silently, either with an all-clear answer or with a duress answer that looks, to anyone watching, exactly like an all-clear answer but summons help. Non-lexical articulatory cues, such as a pattern of tongue presses, a silent tongue click, or a jaw clench, may serve as a safe input for a person who cannot form words.

[0013] In another aspect, the system keeps protection in place when a wearable device is taken off, as eyeglasses often are in bed, by handing the guard state to another sensor of the user, such as earbuds, a neckband, a sleep mask or headband, a ring or button, or a non-imaging sensor near the bed directed at the user's face, and it tells the user privately if no silent channel remains.

[0014] In another aspect, within an affirmative consent envelope in which media plays only while each participant renews an affirmation, a silently articulated affirmation counts only where every participant has elected it, and its effect is made perceptible to the other participant, for example through the volume of the media or a haptic pulse on the other participant's wearable, because consent is communicated between people. A silently articulated stop is honored at once, and its response is perceptible to everyone in the room without revealing who gave it or how.

[0015] In another aspect, a wearable button that opens a voice session while it is held serves as an arming cue, as a safe input through an enrolled press pattern, and as a channel through which a safe phrase or a check-in answer is spoken. The check-in is delivered on another device where the button has no speaker, and the safety recognizer runs on the user's device independently of whatever assistant the button's session reaches.

[0016] In a further aspect, an artificial-intelligence assistant that receives, within a session, a safe phrase or a request for help, whether spoken, typed, or articulated silently, invokes the private check-in and help response of the safety system through an interface that the system exposes, rather than answering conversationally, and passes nothing of the session beyond the invocation. In yet another aspect, the user may say, in the user's own words, what response is wanted, such as a pretext call, a message to a contact, a ride, or help. An utterance is treated as addressed to the system, without a wake word, when it is articulated silently, whispered to a near-field sensor, given while a button is held, or given within a window opened by a check-in; it is interpreted on the device into actions that the user approved in advance; and free speech can never cancel a pending help response.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0017] FIG. 1 is a block diagram of a system 100 in which wearable and nearby sensors of a user's articulators, a microphone, a wearable button, and an artificial-intelligence assistant provide inputs to a guard state and an affirmative consent envelope running on a user device, which presents private check-ins and initiates a tiered help response.

[0018] FIG. 2 is a flow chart of a method 200 by which a guard state detects a safe phrase or other safe input given aloud, in a whisper, or silently, verifies it, presents a private check-in that the user may answer silently, and initiates a tiered help response, including a hand-off of the guard state when a wearable sensor is taken off.

DETAILED DESCRIPTION

[0019] The following description sets out embodiments by way of example. Each element described may be used, omitted, or replaced independently of the others and combined with any element of the related applications, and the words "may," "for example," and "such as" introduce non-limiting examples.

1. Definitions

[0020] "Articulators" means the tongue, lips, jaw, soft palate, and larynx of a person, and the muscles that move them.

[0021] "Speech sensor" means any sensor from whose signal words, phrases, or articulatory cues formed by a person can be recognized. It includes a microphone (an "acoustic speech sensor") and a "non-acoustic speech sensor," which senses the movement, position, or activity of the person's articulators other than by the sound they produce. Non-acoustic speech sensors include: a radio-frequency sensor, in which one or more antennas worn on the head, for example embedded in the frames of eyewear, transmit a low-power radio signal through or across the face and receive it, the transmission, reflection, or coupling of the signal varying with the positions of the tongue, lips, and soft palate; a millimeter-wave or other radar sensor, worn or placed near the person and directed at the face or neck; a surface electromyography sensor on the face, jaw, chin, or neck; an ultrasonic sensor that emits sound above the range of hearing and senses its reflection from the face or the oral cavity; an inertial or vibration sensor on the jaw, cheek, ear, or throat, including a sensor of laryngeal vibration; an ear-canal sensor that senses deformation of the ear canal or changes of pressure in it as the jaw moves; an optical non-imaging sensor, such as a photodiode or proximity sensor, directed at the lips or cheek; a capacitive or pressure sensor in an oral appliance, such as a

night guard or retainer, that senses contact of the tongue or teeth; and, only where the person has expressly enabled it, a camera directed at the lips whose frames are processed on the device and not retained. A non-acoustic speech sensor "worn by" or "directed at" a person senses that person's articulators and, by its placement, substantially no one else's.

[0022] "Silent articulation" means forming a word, a phrase, or an articulatory cue with the articulators while producing no audible sound, or a sound below ordinary conversational audibility. It includes mouthing words with moving lips, articulating with the lips closed so that chiefly the tongue and soft palate move, and whispering. "Silently articulated" has the corresponding meaning. Silent articulation is a deliberate act and is distinguished from involuntary movement of the articulators during thought, which the system does not treat as input.

[0023] "Articulatory cue" means a deliberate, non-lexical movement of the articulators that a person enrolls as an input, such as a pattern of presses of the tongue against the palate or teeth, a silent tongue click, a pattern of taps of the teeth, a jaw clench held for a set time or repeated in a pattern, or a pursing of the lips repeated in a pattern.

[0024] "Safe phrase" means a word or phrase enrolled by a user, or a universal emergency phrase adopted under a policy, whose detection in the guard state starts a private check-in. "Safe input" means a safe phrase or any other input enrolled to have the same effect, including an articulatory cue, a button press pattern, a tap or touch pattern, or a gesture. "Arming cue" means an input by which the guard state is entered, including an ordinary input that does not name a safety function. "Affirmation input" and "stop input" have the meanings given in the '657 application and include inputs given by silent articulation. "Participant" means a person taking part in a session, and "Duo mode" and "Solo mode" refer to sessions in which, respectively, every enrolled participant or only the participant who armed the session takes part in the system.

2. The guard state, check-in, help response, and consent envelope

[0025] Guard state. A user device enters a guard state upon an arming cue. The arming cue may be an ordinary input that does not name a safety function and that the user designated in advance or a default configuration supplies, such as a media selection (a playlist, a track, an artist, or a station), an ordinary request to a voice assistant, a command to a connected device, a press or touch pattern on a button or wearable device, a gesture, a placement of a phone, or a silently articulated phrase enrolled for the purpose; an explicit command naming the guard state may also arm it. The ordinary response to the cue, such as playing the selected media, is carried out as usual, and entering the

guard state produces no output other than that response and, optionally, a private acknowledgement such as a haptic pulse on a device of the user. The guard state may be entered in any room or only while the device is determined to be in the user's bedroom, for example from a smart-home room assignment, a room tag, or a designation learned from where the device rests during the user's sleep. In the guard state no affirmation is awaited, the guard state never implies consent to anything, and it continues until the user who armed it closes it.

[0026] Detection without a wake word. In the guard state the device detects a safe input, including a safe phrase, without first detecting a wake word. By default no audio or other sensor signal is retained, and no transcript is generated of anything other than a matched safe input.

[0027] Private check-in and help response. Upon a safe input, the device presents a private check-in on one or more devices of the user, for example a haptic pulse on a watch or ring, a notification on a phone, audio that only the user hears through an earbud or by bone conduction, or a line on the display of eyewear, and repeats it once with greater intensity if it is not answered. The user answers by a single input. An all-clear answer returns the device to the guard state. An answer requesting help, or a duress answer that the user enrolled and that presents exactly as an all-clear answer, initiates a help response, and so does the absence of any answer. The help response is tiered: it first notifies a contact the user designated, optionally together with a monitoring center that attempts to reach the user, and it requests emergency services upon an answer requesting help, a duress answer, a corroborating distress signal (such as a scream, an impact, a pattern consistent with freezing, or a physiological distress signal from a wearable device), or a failure to reach any designated contact within a configured period. A request to emergency services carries a validation level computed from the evidence, so that a dispatch center can set its response. Media playing in the room may be stopped or, at the user's election, faded as an ambient signal that a check-in is pending.

[0028] Affirmative consent envelope. Where the participants elect it, a session may run within an affirmative consent envelope, in which media continues to play only while each participant renews an affirmation within an affirmation window, and a lapse fades the media and issues a check-in. In Duo mode every enrolled participant must affirm; in Solo mode, the participant who armed the session. The volume of the media may follow, by a linear or nonlinear function, the lowest current consent score among the participants, so that the media is an ambient indication of the score, and a score below a threshold fades the media and issues a check-in. Absent a deliberate affirmative signal through an input that a participant enrolled, that participant's score decays gradually, and an involuntary physiological signal never raises it. A stop input of any participant fades or stops the

media and issues a check-in, and the system announces only its response, never which participant gave the stop or by what input.

[0029] Universal phrase and relay. A universal emergency phrase adopted under a policy for a locale may be recognized by any device. A device not bound to the user that detects it may broadcast over a short-range wireless link an anonymous alert, carrying no identifier of any person or account, which a nearby device of the user treats as a detection of the safe phrase.

3. System overview (FIG. 1)

[0030] Referring to FIG. 1, a system 100 comprises one or more speech sensors worn by or directed at a user, a user device 120, and further devices of the user and of a help pathway 140. The non-acoustic speech sensors include an articulation sensor 104 in the frames of eyewear 102; an earbud 106, which may carry an ear-canal, in-ear radio-frequency, or vibration sensor and which may also deliver audio that only the user hears; a neckband 108, which may take the form of a necklace, collar, or choker and carry a radio-frequency, electromyography, or laryngeal-vibration sensor directed at the throat and the floor of the mouth; a sleep mask or headband 110, which may carry a radio-frequency or electromyography sensor directed at the cheeks and jaw and is suited to wear in bed; and a bedside sensor 112, a non-imaging radio-frequency or radar sensor placed near the bed, for example on a headboard or nightstand, and directed at the user's face, from whose signal no image recognizable as a person can be rendered. An oral appliance with tongue-contact sensing may also serve. A microphone 118 of the user device 120, of a watch 132, of an earbud, or of another device is an acoustic speech sensor and is optional. A wearable button 114, which may be clipped to clothing, worn on a lanyard or band, or adhered to a surface such as a headboard, has one or more buttons supporting press, double-press, hold, and directional-swipe inputs, may be rotated, and has a microphone 116 that is active only while a button is held.

[0031] The user device 120, for example a phone, comprises a processor and memory storing an articulation recognizer 122, a guard-state module 124, a check-in module 126, an envelope module 128, and an assistant interface 130. Each sensor communicates with the user device over a short-range wireless link, such as Bluetooth Low Energy, or runs the recognizer on its own processor and sends only recognized events. The check-in module 126 presents check-ins on the user device, on the watch 132, through the earbud 106, or on a display of the eyewear 102. The envelope module 128 controls a playback device 134, and through it the media in the room, and may signal a wearable device 148 of another participant. The assistant interface 130 allows an artificial-intelligence assistant 136 that the user chooses to invoke the check-in and help response of the

guard state. The help pathway 140 comprises one or more of a designated contact 142, a monitoring center 144, and emergency services 146. The steps described may be performed on any one of these devices or divided among them and a server, and the eyewear 102 or another wearable device may itself serve as the user device.

4. Detecting a safe input given silently (FIG. 2)

[0032] Referring to FIG. 2, a method 200 begins when the guard-state module 124 detects an arming cue 202 and enters the guard state 204, carrying out any ordinary response to the cue. In the guard state, the articulation recognizer 122 monitors 206 the signal of every speech sensor that the user is wearing or that is directed at the user, including any non-acoustic speech sensor and, where enabled, a microphone. It recognizes a safe phrase, without first detecting a wake word, whether the user speaks it aloud, whispers it, or articulates it silently.

[0033] The recognizer may decode the sensor signal into words with a general silent-speech model and compare the decoded words with the enrolled safe phrase, or it may match the signal directly against templates of the user's own renditions of the safe phrase, recorded at enrollment, by an embedding distance or a dedicated keyword-spotting model. The second approach needs no general vocabulary, runs at low power, and works with closed-lip articulation. A safe phrase may be enrolled to be accepted however it is articulated, or as a silent-only phrase that the system ignores when spoken aloud, so that the user can say an ordinary word aloud in conversation while reserving its silent articulation as the safe phrase.

[0034] Where several speech sensors are present, their evidence is fused. A candidate detected in the signal of one sensor above that sensor's threshold suffices, and agreement among sensors, as when both a microphone and an articulation sensor register a whispered phrase, raises the confidence. Because a non-acoustic speech sensor senses only its wearer, a detection from it is attributed to the wearer without voice verification, cannot be produced by another person speaking, and is unaffected by media, so that no cancellation of media is required for it. A candidate detected only by a microphone remains subject to the measures of the related applications for distinguishing live speech from media, such as cancellation, lyric matching, and verification of the user's enrolled voice.

[0035] A candidate safe input 208 passes to verification 210, which accepts it when its confidence exceeds a threshold and, preferably, when it is repeated within a short interval or corroborated by another sensor, and which accepts rather than rejects a candidate whose classification is uncertain,

the check-in itself guarding against a false alarm. An accepted safe input leads to the private check-in 212 described in section 7; a rejected candidate returns the method to monitoring 206.

5. Articulatory cues for a person who cannot form words

[0036] A person who is frozen, held, or struggling to breathe may be unable to form even silent words. The user may therefore enroll an articulatory cue as a safe input, such as three firm presses of the tongue against the palate, a silent tongue click repeated twice, a jaw clench held for about two seconds, or a pattern of teeth taps, each chosen to be distinct from chewing, swallowing, kissing, and other ordinary movements. An articulatory cue may also serve as an arming cue, as an answer to a check-in, or as an affirmation or stop input. Over the user's ordinary wear, the system learns which movements the user makes involuntarily and prompts the user to choose a different cue where an enrolled cue resembles one of them.

[0037] A freeze response may itself be sensed. A sudden cessation of articulatory and breathing movement after a safe input, or a pattern of stillness, held breath, and elevated heart rate, may serve as a corroborating distress signal in the tiered help response. Such a pattern alone never summons emergency services, and it never counts as an affirmation of anything.

6. Enrollment, rehearsal, and calibration

[0038] At enrollment, the user articulates each safe phrase, check-in answer, and articulatory cue silently two or three times, and the system records templates on the device. A short calibration, for example reading a few sentences silently, adapts a general model to the user's anatomy and to the fit of the device. A labeled rehearsal mode lets the user practice giving each input silently, in a whisper, aloud, and while lying down, and feel or hear the check-in, while help pathways are simulated rather than dispatched. Because articulation signals change with the fit of eyewear, the position of the head on a pillow, or a change of device, the system checks recognition periodically with the user's consent and asks for a fresh rendition when its confidence drifts.

7. Answering a check-in silently

[0039] The private check-in 212 is delivered where only the user perceives it: a haptic pulse on the watch 132 or a ring, a soft voice through the earbud 106 or by bone conduction from the eyewear 102, or a line on the display of the eyewear 102. The user answers 214 by silent articulation, aloud, or by any other enrolled input. A silently articulated all-clear answer 216 returns the method to the guard state. A silently articulated duress answer 218, which the user enrolled and which the system treats exactly as an all-clear answer in everything that anyone can perceive, initiates the help

response 220, so that a person who is watched while answering can summon help without any visible sign; an answer requesting help does the same. Absence of an answer 222 after the repeated check-in initiates the tiered help response 220. At any time in the guard state, the user may also articulate silently a request concerning a designated contact, such as asking that the contact be messaged, which the system carries out privately.

8. Privacy and data handling

[0040] A non-acoustic speech sensor senses only its wearer, so a guard state that uses it captures no speech of any other person and may run with every microphone in the room closed. That matters where the law of a place requires every party's consent to a recording, and it reassures a partner that nothing said in the room is being heard. Articulation signals are processed on the wearable device or the user device; unless the user elects otherwise, they are not sent to a server, are not retained after recognition, and produce no transcript other than of a matched input. Because such signals can reveal words that a person did not intend to share, the system gives them the protections that the '657 application applies to neural and biometric data: processing on the device, no secondary use, no sale, and deletion on request. A radio-frequency sensor operates at low power within applicable exposure limits.

9. Sensor placements, and hand-off when a device is taken off

[0041] Eyeglasses are often taken off before sleep or intimacy, so the guard state does not depend on any one wearable device. Each wearable device detects whether it is worn, for example from the coupling of its antennas to the face, from skin contact, or from a proximity sensor. When the sensor that the guard state is using is taken off 224, the guard-state module 124 hands the guard state to the remaining channels: another non-acoustic speech sensor that the user wears, such as the earbud 106, the neckband 108, or the sleep mask or headband 110; the bedside sensor 112; a ring, watch, or wearable button 114 on which a press, tap, or squeeze pattern serves as the safe input; or, where enabled, a microphone. If no silent channel remains, the user is told privately, for example by a haptic pulse, and may choose a remaining channel or accept the microphone. The hand-off is silent and visible to no one else.

[0042] A non-acoustic speech sensor may be placed in any wearable device that rests near the articulators: the temples, hinges, nose pads, or bridge of eyewear; an earbud, a hearing aid, or the stem of an earbud; a band behind the ear; a necklace, collar, choker, or neckband; a headband, cap, or sleep mask; a chin strap; a mouthguard, night guard, or retainer; a face mask; or a patch on the

skin of the jaw or throat. Several placements may be used together, and after calibration the same recognizer serves all of them.

10. Sensing from beside the bed

[0043] The bedside sensor 112 is a non-imaging radio-frequency or radar sensor that senses the motion of the lips, jaw, and throat of a person lying in a known region of the bed, with nothing worn. It directs its beams toward the region where the user's head lies, learned from where the user sleeps, and recognizes a safe phrase or articulatory cue articulated silently there. Because a bedside sensor may sense more than one person, it attributes a detection to the user by the position of its source, by the user's enrolled articulation pattern, or by corroboration from a device that the user wears, and it does not reconstruct any image. Where two participants each enroll, each person's safe input is attributed to that person.

11. Silent signals in the affirmative consent envelope

[0044] Consent is communicated between people, and a signal that only a device perceives communicates nothing to the other participant. A silently articulated go word, or any other affirmation given by silent articulation, therefore counts toward the affirmative consent envelope only where every participant has elected to allow silent affirmations, and its effect is made perceptible to the other participant: through the volume of the media, which follows the consent score as described in section 2; through a light; or through a haptic pulse on the wearable device 148 of the other participant indicating that an affirmation was received, without revealing its content. A silent affirmation renews the participant's affirmation window and arrests the gradual decay of that participant's score, in the same way as a squeeze, a tap pattern, or another deliberate non-verbal affirmation. Unless every participant has elected otherwise, it never by itself raises the score above the grade that the participant's spoken or otherwise perceptible affirmations have established.

[0045] A silently articulated stop input, by contrast, is always honored at once and needs no one's election, because a stop must never depend on being heard. Upon it, the media fades or stops and a check-in issues, perceptibly to every participant, and the system announces only its response, never which participant gave the stop or by what input. A participant who cannot say stop aloud can therefore stop an activity, and the other participant learns that it has stopped without learning how.

[0046] A participant may also answer the envelope's check-ins silently, and may enroll a silent concern cue (a grey-zone, yellow-flag, or brake signal, in the terminology of the '657 application) that raises the corresponding response without the other participant hearing anything.

12. A wearable button that opens a voice session while held

[0047] The wearable button 114 may be a general-purpose voice remote whose ordinary function is to open a voice session with an assistant, take notes, or control connected devices. Any of its ordinary inputs may be enrolled as an arming cue, such as a double press, or a hold during which the user asks the assistant an ordinary question, so that arming looks like ordinary use. In the guard state, a press pattern enrolled as a safe input, such as a hold combined with a swipe in a particular direction, or three presses, starts the private check-in. While a button is held and its microphone 116 is active, the captured speech is passed to the articulation recognizer 122 on the user device before, or in parallel with, its forwarding to any assistant, so that a safe phrase spoken in that session starts the check-in whatever the assistant does with the session. The button's own ordinary response, such as a confirming beep, is unchanged, so that nothing perceptible distinguishes a safe input from ordinary use.

[0048] The button may have no speaker or display, in which case the check-in is delivered on the user device 120, the watch 132, or the earbud 106, and it is answered by a press pattern on the button, by holding the button and answering aloud or in a whisper, or by any other enrolled input. A button adhered to a headboard or nightstand, or clipped within reach of the bed, offers a safe input that needs neither speech nor a wearable device. The button may also be rotated or swiped to control the volume of the media in the room in the ordinary way, and that use is not interpreted as a safety signal unless enrolled as one.

[0049] Because the safety recognizer runs on the user device, the safety function depends neither on the assistant that the button reaches, nor on that assistant's provider, nor on a network connection to it.

13. An interface for artificial-intelligence assistants

[0050] Many people now speak, type, or silently articulate to an artificial-intelligence assistant 136, including through a silent-speech input device or a held-to-talk button. The assistant interface 130 exposes to an assistant that the user has authorized a narrow function, such as a tool, an intent, or a webhook, by which the assistant invokes the guard state's private check-in and help response. When an assistant recognizes within a session that the user has given a safe phrase, a universal emergency

phrase, or a request for help, it invokes the function rather than answering conversationally or advising the user to call for help, and it passes nothing of the session beyond the invocation and, optionally, a category of the request. The system then runs its own private check-in and tiered help response with its own verification, so that the assistant cannot summon emergency services by itself and the help response does not depend on the assistant's judgment. Where the guard state is active, an invocation from an assistant is treated as a safe input. Where it is not, the system presents a private check-in offering help and, unless the user has configured otherwise, initiates no help response upon absence of an answer.

[0051] The interface also works in the other direction. An ordinary request that the user makes to an assistant may be enrolled as an arming cue, and while the guard state is active the assistant may be told only that a safety session is active, so that it does not respond to the safe phrase in a way that reveals it, such as by repeating it aloud.

14. Silent inputs outside the bedroom

[0052] The same silent inputs serve the other mechanisms of the related applications. A user who arrives at an unfamiliar place late at night may answer the arrival check-in of the '542 application silently; a user may request refuge for the night by silent articulation without anyone nearby hearing the request; and a user in a vehicle, at a party, or on a date may give a silent safe phrase that starts the same private check-in. A silently articulated universal emergency phrase is detected by the user's own sensor, so it starts the user's own check-in without any broadcast.

15. Limiting false alarms

[0053] Silent articulation can occur in conversation, in sleep, and in reading, and a non-acoustic sensor can produce spurious candidates. The measures of the related applications therefore apply: a confidence threshold, preferably with repetition or corroboration; a check-in delivered at once on several devices and repeated once; a tiered help response that notifies a designated contact before emergency services and requests emergency services only upon an answer requesting help, a duress answer, a corroborating distress signal, or a failure to reach any contact; and a validation level carried with any request to emergency services. While the user sleeps, the guard state may require a silent safe phrase to be repeated, or accept only an articulatory cue that does not occur in sleep. The system learns from detections that the user dismisses as mistaken and prompts the user to choose a less ambiguous phrase, and it never lowers protection after a detection that the user did not dismiss. The guard state ends 226 only when the user who armed it closes it.

16. Variations

[0054] Wherever this disclosure or the related applications describe detecting a word, a phrase, or a vocal cue in audio captured by a microphone, the detection may equally be performed on the signal of any speech sensor, acoustic or non-acoustic, or of several in combination, and the word, phrase, or cue may be given aloud, in a whisper, or by silent articulation. Every input described here may serve as an arming cue, a safe input, an answer to a check-in, an affirmation input, or a stop input, as the user enrolls it. The steps may be performed by a single device or divided among the user's wearable devices, phone, and other devices and a server. A sensor may come from the manufacturer of the user device or from another manufacturer, reporting recognized events through an interface, and the system may accept events from any sensor that meets its privacy requirements.

17. Instructions in the user's own words

[0055] Beyond enrolled phrases and codes. A user in the guard state may also say, in the user's own words and in any language, what response the user wants, rather than giving an enrolled safe phrase or code: for example, "have Sam call me with an emergency in ten minutes," "text my sister where I am and that I need a ride," "get me a car home," "just fade the music and check on both of us," "check on me in half an hour," or "call the police." An on-device language model, of the kind that the '657 application uses to classify the meaning of utterances for consent, interprets the instruction into a plan of one or more actions drawn from a set that the user approved in advance, and the system carries out the plan privately.

[0056] Knowing that an instruction is addressed to the system. Because the guard state awaits no wake word, the system must tell an instruction meant for it from ordinary conversation with a partner, which it does not interpret. An utterance is treated as addressed to the system when it is articulated silently, because people rarely mouth words silently to a partner who is present; when it is whispered and sensed by a near-field speech sensor of the user, such as the articulation sensor 104, the earbud 106, or the neckband 108, below the level at which another person in the room would hear it; when it is spoken while the user holds the wearable button 114; when it is given within an instruction window that opens upon a safe input, a private check-in 212, or an enrolled tap, and that the user perceives privately, for example as a haptic pulse; or when it begins with a name for the system that the user chose and that sounds ordinary in conversation. Outside these conditions, speech is not transcribed for this purpose, consistent with section 8.

[0057] The action set. The actions that an instruction may select include: a pretext call or message to the user, at a time the user names, from a contact the user names, or from the system in that contact's name where the contact has agreed; a message to a designated contact, with or without the user's location, such as a request to come or to call; sharing the user's location with a designated contact for a period; a timed check-in, such as one in thirty minutes; a ride to the user's home or to a refuge, or a request for a refuge for the night under the '542 application; fading, stopping, or resuming media; a check-in for every participant; arming or closing the bedroom mode or the affirmative consent envelope; starting a recording where the user's settings and the law allow it; the tiered help response; and a request to emergency services. The user approves the action set at enrollment and may restrict it. An action outside the set is not performed, and the user is told so privately.

[0058] Ambiguity and confirmation. Where an instruction is ambiguous, the system resolves the ambiguity toward safety: it presents a private check-in offering the most likely actions, answerable by a single input, rather than guessing. A pretext call or message is scheduled as asked, without confirmation, because a request for a way out should meet no friction; an action that would summon another person to the place, other than a request for help, may be confirmed privately by a single input. An instruction that expresses distress or a wish for help, however worded, starts the private check-in 212 and, upon the user's request or absence of an answer, the tiered help response 220.

[0059] Free speech cannot cancel help. A person may be compelled to say "I'm fine," "cancel that," or "never mind." An instruction in the user's own words may therefore add to, hasten, or choose the form of a help response, but it never cancels, delays, or downgrades a help response or check-in that is pending; only the enrolled all-clear answer does so, and the enrolled duress answer remains available. Likewise, an instruction never closes the guard state, disables a sensor, or deletes the user's records; those actions require the user's enrolled inputs, or the settings interface at a time when no check-in is pending.

[0060] Attribution and privacy. Only instructions attributed to the user, by a non-acoustic speech sensor that the user wears, by the wearable button 114, or by verification of the user's enrolled voice, are interpreted; instructions in any other voice are ignored, so that another person cannot direct the user's safety system. Interpretation runs on the device. Unless the user elects otherwise, the user's sealed record stores the actions taken and their times, not a transcript of the instruction, and nothing the system does reveals the instruction or the user's input to another person unless the action itself requires it.

[0061] Instructions about how the system responds. The user may also tell the system, in the user's own words, how to respond to the consent signals that the '657 application interprets, for example "ask him to check in with me," "slow it down gently," "don't stop the music, just remind us," or "tell me privately when we're near the end of the window." The system maps such an instruction onto the response options of the active policy, such as the media response on a lapse, the form and addressee of a prompt, or the timing of a reminder, and applies it for the session or, at the user's election, saves it as a preference. In Duo mode, a change affecting the other participant is applied only after that participant accepts it, except that an instruction may always make the response more protective of the person who gave it. No instruction turns a stop into anything less than a stop.

[0062] The same instructions may be given to an artificial-intelligence assistant 136 that the user chose, which passes the interpreted plan through the assistant interface 130, the safety system applying the rules of this section in place of the assistant's own judgment.

EXEMPLARY AND ILLUSTRATIVE CLAIMS

The following claims are exemplary and illustrative. They are presented to illustrate the subject matter disclosed and do not limit the scope of this disclosure or of any claims that may later be presented in an application claiming the benefit of this provisional application.

1. A method performed by one or more computing devices including a user device, the method comprising: entering a guard state responsive to an arming cue given by a user; in the guard state, detecting, without first detecting a wake word, a safe phrase articulated by the user, from a signal of a non-acoustic speech sensor that is worn by or directed at the user and that senses movement or activity of articulators of the user, the safe phrase being detectable when articulated silently; and, upon detecting the safe phrase, presenting a private check-in on a device of the user and, upon absence of a response to the private check-in, initiating a help response.

2. The method of claim 1, wherein the non-acoustic speech sensor comprises one or more antennas worn on the head of the user that transmit a radio-frequency signal through or across the face of the user and receive it, the safe phrase being detected from variation of the received signal with positions of at least one of the tongue, the lips, and the soft palate of the user.

3. The method of claim 2, wherein the antennas are embedded in frames of eyewear worn by the user.

4. The method of claim 1, wherein the non-acoustic speech sensor is selected from a radar sensor, a surface electromyography sensor, an ultrasonic sensor that emits inaudible sound and senses its reflection, an inertial or vibration sensor, an ear-canal sensor, an optical non-imaging sensor, and a contact sensor of an oral appliance, and is carried by at least one of eyewear, an earbud or hearing aid, a neckband, collar, or necklace, a headband, cap, or sleep mask, a chin strap, an oral appliance, and a patch on the skin.

5. The method of claim 1, wherein the safe phrase is detected when articulated with the lips closed.

6. The method of claim 1, wherein the arming cue is an ordinary input that does not name a safety function, designated in advance by the user or by a default configuration, and selected from a media selection, a request to a voice assistant, a command to a connected device, a button or touch pattern, a gesture, a placement of a device, and a phrase articulated silently, and wherein any ordinary response to the arming cue is carried out.

7. The method of claim 1, wherein the safe phrase is detected while media plays within range of the user without cancelling the media from any signal, a detection from the non-acoustic speech sensor being attributed to the user without voice verification.

8. The method of claim 1, wherein the guard state is maintained while no microphone in a room of the user is capturing audio.

9. The method of claim 1, wherein the safe phrase is enrolled as a silent-only safe phrase that is not acted upon when spoken aloud.

10. The method of claim 1, further comprising detecting, from signals of a microphone and of the non-acoustic speech sensor, a whispered or spoken rendition of the safe phrase, a detection in either signal above a respective threshold sufficing, and agreement between the signals raising a confidence of the detection.

11. The method of claim 1, further comprising detecting, from the signal of the non-acoustic speech sensor, an articulatory cue enrolled by the user and selected from a pattern of presses of the tongue, a silent tongue click, a pattern of taps of the teeth, and a jaw clench held for a set time or repeated in a pattern, and treating the articulatory cue as the safe phrase.

12. The method of claim 1, wherein the private check-in is delivered by at least one of a haptic pulse on a wearable device, audio perceptible only to the user through an earbud or by bone conduction, and a display of eyewear, and wherein the user answers the private check-in by silent articulation with either an all-clear answer, which returns to the guard state, or an enrolled duress answer, which is treated as the all-clear answer in everything perceptible and initiates the help response.

13. The method of claim 1, further comprising detecting that a wearable device carrying the non-acoustic speech sensor has been taken off and, responsive thereto, continuing the guard state with at least one remaining channel selected from another non-acoustic speech sensor worn by the user, a non-imaging sensor directed at the user from beside a bed, a press, tap, or squeeze pattern on a wearable device or button, and a microphone, and privately notifying the user where no channel for silent input remains.

14. The method of claim 1, wherein the non-acoustic speech sensor is a non-imaging radio-frequency or radar sensor placed beside a bed and directed at a region where the head of the user lies, from whose signal no image recognizable as a person is rendered, a detection being attributed

to the user by at least one of a position of its source, an enrolled articulation pattern of the user, and corroboration from a device worn by the user.

15. The method of claim 1, wherein the help response is tiered, first notifying a contact designated by the user and requesting emergency services only upon at least one of an answer requesting help, a duress answer, a corroborating distress signal, and a failure to reach any designated contact within a configured period.

16. A method performed by one or more computing devices, the method comprising: playing media in a room within an affirmative consent envelope in which the media continues to play only while each of a plurality of participants renews an affirmation within an affirmation window; receiving an affirmation of a first participant as a go word articulated silently by the first participant and detected from a signal of a non-acoustic speech sensor worn by or directed at the first participant, the affirmation being counted only where each participant has elected to allow silent affirmations; making an effect of the affirmation perceptible to a second participant through at least one of a volume of the media, a light, and a haptic pulse on a wearable device of the second participant, without revealing the content of the affirmation; and, upon a lapse of the affirmation window of any participant, fading the media and issuing a check-in.

17. The method of claim 16, further comprising, upon a stop input articulated silently by any participant and detected from a signal of a non-acoustic speech sensor, fading or stopping the media and issuing a check-in perceptibly to every participant without indicating which participant gave the stop input or by what input.

18. The method of claim 16, wherein a volume of the media follows, by a linear or nonlinear function, a lowest current consent score among the participants, and wherein a silent affirmation arrests a gradual decay of the consent score of the participant who gave it without raising that consent score above a grade established by perceptible affirmations of that participant.

19. A method performed by one or more computing devices including a user device, the method comprising: receiving, from a wearable button device having a microphone that is active only while a button of the wearable button device is held, button inputs and audio captured while the button is held; entering a guard state upon an arming cue comprising an ordinary input of the wearable button device; in the guard state, detecting on the user device a safe input selected from a press pattern of the button enrolled by a user and a safe phrase in the audio captured while the button is held, the audio being analyzed on the user device before or in parallel with forwarding of the audio to an assistant; and, upon detecting the safe input, presenting a private check-in on a device of the user

other than the wearable button device and, upon absence of a response to the private check-in, initiating a help response.

20. The method of claim 19, wherein the wearable button device produces the same perceptible response to the safe input as to an ordinary input.

21. A method performed by one or more computing devices, the method comprising: exposing, by a safety system on a user device, to an artificial-intelligence assistant authorized by a user, a function for invoking a private check-in and a help response of the safety system; recognizing, by the assistant within a session with the user in which input of the user is spoken, typed, or articulated silently, a safe phrase, a universal emergency phrase, or a request for help; invoking, by the assistant, the function in place of answering conversationally, the invocation conveying no content of the session other than, optionally, a category of the request; and, responsive to the invocation, presenting by the safety system a private check-in on a device of the user and initiating the help response according to an answer to the private check-in, whereby the assistant does not itself request emergency services.

22. The method of claim 21, wherein, where no guard state is active on the user device, the safety system initiates no help response upon absence of an answer to the private check-in.

23. A system comprising a non-acoustic speech sensor configured to be worn by or directed at a user and to sense movement or activity of articulators of the user, and a user device comprising a processor and memory storing instructions that, when executed by the processor, cause the system to perform the method of any one of claims 1 to 15 and 25 to 27.

24. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of any one of claims 1 to 22 and 25 to 28.

25. A method performed by one or more computing devices including a user device, the method comprising: in a guard state entered responsive to an arming cue, and without first detecting a wake word, determining that an utterance of a user is addressed to the user device by at least one of the utterance being articulated silently and sensed by a non-acoustic speech sensor worn by or directed at the user, the utterance being whispered and sensed by a near-field speech sensor worn by the user, the utterance being given while the user holds a button of a wearable device, and the utterance being given within an instruction window that is opened by a safe input or a private check-in and indicated privately to the user; interpreting, on the user device, the utterance as an instruction in the

user's own words into a plan of one or more actions selected from a set of actions approved by the user in advance; and carrying out the plan, wherein utterances not determined to be addressed to the user device are not interpreted.

26. The method of claim 25, wherein the set of actions comprises at least two of a pretext call or message to the user at a time named in the instruction, a message to a designated contact with or without a location of the user, sharing the location of the user with a designated contact for a period, a timed check-in, a ride or a request for refuge, fading or stopping media, a check-in for every participant, a tiered help response, and a request to emergency services, and wherein an ambiguous instruction results in a private check-in offering the most likely actions.

27. The method of claim 25, wherein an instruction may add to, hasten, or select a form of a help response but does not cancel, delay, or downgrade a pending help response or check-in, which only an enrolled all-clear answer does, and wherein an instruction does not close the guard state.

28. A method performed by one or more computing devices, the method comprising: running a session in which media plays within an affirmative consent envelope and consent signals of participants are interpreted into responses according to an active policy; receiving from a participant an instruction, in the participant's own words, about how the system is to respond; mapping the instruction onto response options of the active policy, including at least one of a media response on a lapse, a form and addressee of a prompt, and a timing of a reminder; and applying the mapped response, wherein a change affecting another participant is applied only upon acceptance by that participant unless the change makes the response more protective of the participant who gave the instruction, and wherein no instruction reduces a stop to less than a stop.

ABSTRACT OF THE DISCLOSURE

A guard state, armed by an ordinary cue such as a song or a button pattern, detects without a wake word a safe phrase that a user speaks, whispers, or articulates silently, sensed by a non-acoustic sensor of the user's articulators, such as radio-frequency antennas in eyewear frames. Because the sensor senses only its wearer, songs and other voices cannot trigger it, no room microphone is needed, and the other person hears nothing. A detection starts a private check-in, answerable silently with an all-clear or duress answer, before a tiered help response. Removing the eyewear hands the guard state to earbuds, a sleep mask, a bedside radar, or a button. Silent affirmations count toward a consent envelope only when every participant elects them and their effect is perceptible; silent stops act at once. Users may also request responses in their own words, though free speech never cancels pending help.